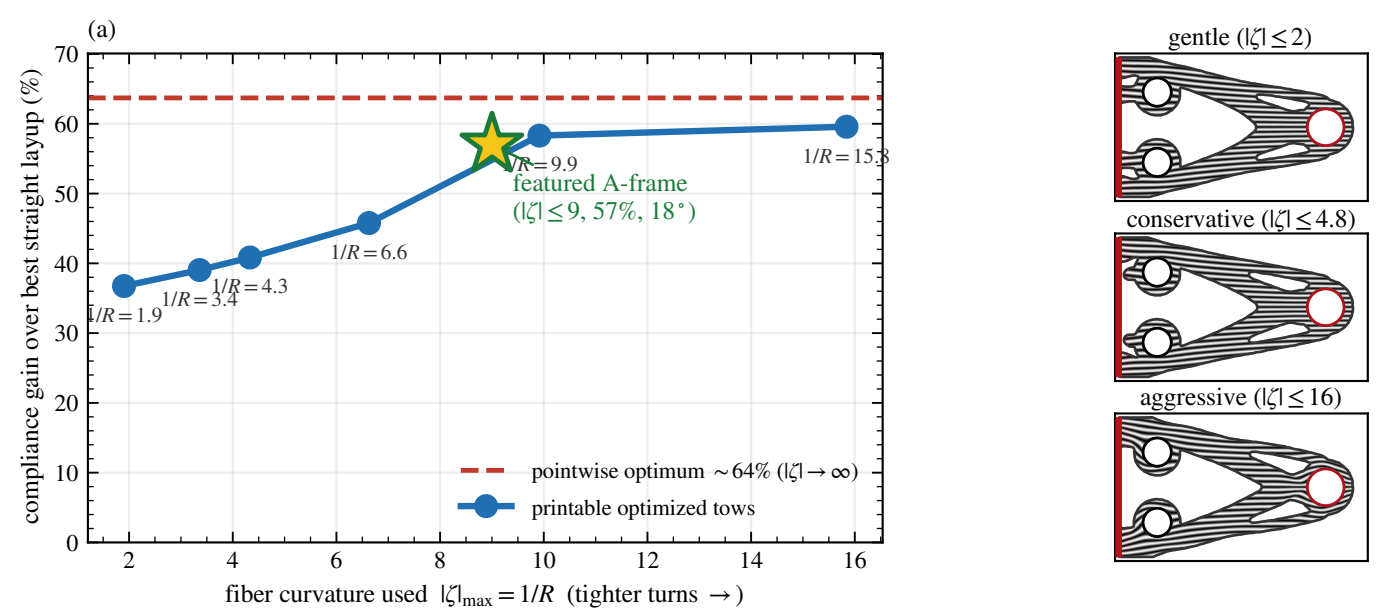

(a)
compliance gain over best straight layup (%)
fiber curvature used $|\zeta|_{\max} = 1/R$ (tighter turns →)
1/R = 1.9
1/R = 3.4
1/R = 4.3
1/R = 6.6
1/R = 9.9
1/R = 15.3
featured A-frame
(|ζ| ≤ 9, 57%, 18°)
pointwise optimum ~64% (|ζ| → ∞)
printable optimized tows
gentle (|ζ| ≤ 2)
conservative (|ζ| ≤ 4.8)
aggressive (|ζ| ≤ 16)